MIL-F-51109A(EA)
20 January 1977
SUPERSEDING
MIL-F-51109(MU)
19 April 1968

MILITARY SPECIFICATION

FACEBLANK, CHEMICAL-BIOLOGICAL MASK, C12R4

This specification is approved for use by Edgewood Arsenal, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a chemical-biological mask faceblank molded from a 65 percent natural rubber compound.

1.2 Classification. The faceblank shall be of the following sizes, as specified (see 6.2):

Small
Medium

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-F-51110 - Faceblank - Lens, C14R3.

FSC 4240

: Beneficial comments (recommendations, additions, deletions) and any :
: pertinent data which may be of use in improving this document should :
: be addressed to: Commander, Edgewood Arsenal, Attn: SAREA-TS-S, :
: Aberdeen Proving Ground, MD 21010 by using the self-addressed :
: Standardization Document Improvement Proposal (DD Form 1426) appear- :
: ing at the end of this document or by letter. :

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STANDARDS

FEDERAL

FED-STD-123 - Marking for Shipping (Civil Agencies).

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

US ARMY ARMAMENT COMMAND

EDGEWOOD ARSENAL

E5-2-875 - Faceblank, Small.
E5-2-806 - Faceblank, Medium.
D136-19-14 - Puncture, Tester, Natural Rubber, Q38, Assembly.
E136-19-22 - Flexibility, Tester, Natural Rubber, Q37, Assembly.

PUBLICATIONS

EDGEWOOD ARSENAL DIRECTIVES

No. 618 - Ozone Resistance Test of Natural Rubber Material.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Standards:

D412 - Method of Tension Testing of Vulcanized Rubber.
D572 - Accelerated Aging of Vulcanized Rubber by the Oxygen Pressure Methods.
D573 - Accelerated Aging of Vulcanized Rubber by Oven Method.
D624 - Test for Tear Resistance of Vulcanized Rubber.

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D2240 - Indentation Hardness of Rubber and Plastic by Means of a Durometer.
D3182 - Standard Materials, Equipment, and Procedures for Mixing Rubber Compounds and for Preparing Vulcanized Test Sheets.

(Applications for copies of the above publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa.)

SCHWARTZ, TULIPAN, AND BIRMINGHAM

Occupational Diseases of the Skin

(This publication may be obtained from the publishers: Lea and Febiger, Philadelphia, Pennsylvania.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 Rubber compound. The rubber compound shall be non-blooming, except for slight wax-type bloom, and shall be compounded with no less than 65 percent, by volume, of new natural rubber. The compound shall be free from lead or mineral sulfates.

3.1.2 Antiozonide. The rubber compound shall contain an antiozonide.

3.1.3 Copper and manganese inhibitor. The rubber compound shall contain a copper and manganese inhibitor.

3.1.4 Color. The rubber compound shall be black.

3.1.5 Toxicity. When the faceblank is worn as a component of a chemical-biological mask, it shall not cause undue sensitization or primary irritation of the wearer's skin. Objective evidence shall be submitted in accordance with 4.1.2.2. Once the rubber formulation has been approved by the Government, the supplier shall not make a major change (6.4) in the formulation or in the curing process.

3.2 Manufacture.

3.2.1 Faceblank. The faceblank shall be manufactured, in one piece, by the conventional process of molding under heat and pressure. Each size specified in the contract or order shall be manufactured in accordance with Drawing E5-2-875 or Drawing E5-2-806 as applicable, in molds approved by Edgewood Arsenal, Aberdeen Proving Ground, Maryland 21010.

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3.2.2 Test slabs. Test slabs shall be fabricated from stock representative of the formulation used for the preproduction sample and from each sublot during regular production of faceblanks in accordance with ASTM 3182. The slabs shall have a cure equivalent to that of the regular production blank (see 6.3).

3.3 Physical and chemical properties. The faceblanks and test slabs shall conform to the requirements listed in table I and 3.5, when tested as specified in table II.

Table I. Physical and chemical properties

Property	Units	Unaged		Aged	
		Minimum	Maximim	Minimum	Maximum
Tensile strength	MPa	20.7		17.6	
Ultimate elongation	%	600	750		20% increase 25% decrease
*Tensile stress at 200% elongation	MPa	1.72	3.10	1.72	3.10
Tear resistance	kN/M				
(Die A or Die B)		52.5			30% decrease
(Die C)		39.4			30% decrease
Tension set at 75% elongation	%		20		
Durometer hardness (Shore "A")		44	50		
Ozone resistance	Minutes	80			
Flexibility					
At 24° ± 3°C.	Grid lines	7.5			
At -32° ± 3°C.	Grid lines	4.8			
Puncture resistance hydrostatic pressure	kPa	65.5			

*Individual specimens at 200% elongation shall have a minimum tensile stress of 1.65 MPa.

3.3.1 Weathering. There shall be no visual discoloration of lenses when assembled into faceblanks and tested for weathering in accordance with MIL-F-51110.

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3.4 Preproduction sample. Prior to the start of regular production, 30 (20 trimmed and 10 untrimmed) sample faceblanks from each mold cavity and 16 test slabs shall be produced using the same methods, materials, equipment, and rubber formulation as will be used during regular production. Any change in formulation, operating conditions, or molds shall require a new preproduction sample.

3.5 Resistance to oxygen-pressure aging. The faceblank shall show no cracking or increased hardness or tackiness after 72 hours of accelerated aging in oxygen, at a pressure 2.10 ± 0.10 megapascals and a temperature of 70° ± 1° C, as specified in table II.

3.6 Workmanship. The faceblank inner and outer surfaces shall be smooth and free from imperfections such as pits, bumps, porous areas, cracks, checks, flash, abrasions, or blisters. The rubber shall be compact and free from holes, bubbles, and foreign matter. Ridges of mold parting lines on outer surfaces shall not be considered as flash when closely trimmed. Faceblanks with small flaws in noncritical areas may be repaired provided there is no change in the formulation or surface continuity. The peripheral sealing area contacting the wearer's skin, the lens sealing surface, the inner surface area of the air inlet an outlet stems, and the inner surface area of the microphone cable stem shall be considered as critical areas of the faceblank. Rips, tears, or punctures shall be considered as damage.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence.

4.1.2.1 Materials. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1, 3.3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.1.2.2 Toxicity. After date of award of contract, and within the time specified in the contract, and without additional cost to the Government,

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the contractor shall submit the information listed below which will constitute his evidence for establishing the nontoxicity of the formulation. The U.S. Army Surgeon General will act as advisor to the Contracting Officer in evaluating this data for determining if the formulation and parts are acceptable. The information relating to the description of the candidate formulations may be classified as "company confidential" and this information will be so treated by the Government.

a. Information relating to candidate formulations:

(1) The name of the contractor and the producing subcontractor, if any.

(2) Description of the item and end item and copies of applicable specifications shall be included.

(3) The number of items under procurement for which the proposed formulation shall be used.

(4) A description of the candidate formulation shall include:

(a) The chemical and trade names of the ingredients.

(b) The quantity of each ingredient, including the range of variation that can be tolerated without loss of quality in the finished item. Quantities shall be expressed in percent by weight and parts by weight.

(c) The proposed mold lubricants, dusting powders, wet spray or other materials used in the manufacturing process.

(d) Proposed cure times and temperatures.

(e) Results of previous toxicity evaluation demonstrating acceptability or evidence of prior Government approval of the proposed formulation.

(f) Information and data to substantiate the nontoxicity of the product. This information will include a complete description of any programs the contractor has carried out to prove nontoxicity, such as prior patch testing with unlined items the use of which involves direct contact with the skin; evidence that the proposed formulation is the same as that used in commercial products; and information on the nature, extent, and length of time of his commercial experience with the formulation. Other evidence may include references to technical or other scientific publication, commercial laboratory reports, opinions or statements of recognized authorities and agencies in the field of dermatology, and any other pertinent data.

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b. Patch testing. If the above information is not sufficient to establish nontoxicity of the rubber formulation, the contractor shall perform 200 patch tests under the supervision of competent medical authorities in the field of dermatology. These patch tests shall be at no additional cost to the Government and shall be conducted in accordance with the prophetic patch test method described in "Occupational Diseases of the Skin" by Schwartz, Tulipan, and Birmingham. A control rubber for use in the patch tests will be furnished by the Government upon written request of the contractor. The patches composed of the candidate rubber formulation shall not produce positive reactions in excess of that of the control rubber patches. A copy of the test results and the evaluation thereof by the medical authorities described above shall be furnished to the contracting officer.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Examination. Fifteen of the sample faceblanks shall be inspected for all requirements of the applicable drawings and this specification.

4.3.2 Tests. Fifteen faceblanks and 16 test slabs shall be tested in accordance with 4.4.4.3 (see table II).

4.3.3 Acceptance/rejection criteria. The preproduction sample shall meet the examinations and tests specified by 4.3.1 and 4.3.2 to be acceptable. The supplier shall obtain written approval, including mold approval, from the contracting officer before proceding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot of faceblanks shall consist of those produced in sequence during one month as indicated by the dots in the medallion on the faceblank. A sublot shall consist of the items produced from one batch of material (see 6.9). A new lot shall be established following the occurence of any event affecting the homogeneity of the lot or use of an alternative approved formulation.

4.4.2 Sampling.

4.4.2.1 For nondestructive examination. Sampling shall be conducted in accordance with MIL-STD-105.

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4.4.2.2 For destructive tests. Fifteen faceblanks and 16 test slabs shall be selected at random from each sublot for testing.

4.4.3 Inspection procedure.

4.4.3.1 For examination and nondestructive tests. The sample faceblanks from the regular production lot shall be examined in accordance with classification of defects and MIL-STD-105.

4.4.3.2 For destructive tests. Sample faceblanks and test slabs shall be tested in accordance with 4.4.4.3 (see table II).

4.4.3.3 Classification of defects.

(a) Faceblank (Dwg E5-2-875 or E5-2-806 as applicable).

Categories		Defects
Critical:		None defined
Major:	AQL	0.065 percent defective
101		Imperfection or damage in critical areas (see 3.6)
	AQL	1.0 percent defective
102		Rubber thickness incorrect
103		Porosity or nonfills
104		Deformation or permanent set
105		Flow cracks which enlarge on probing
106		Contamination (grease or oil)
107		Marking incorrect
108		Damage (other than 101)
109		Split at register
110		Spectacle receptacle holes missing or incorrect
Minor:	AQL	2.5 percent defective
201		Imperfections (other than 101)
202		Excess flash or improper trim
203		Marking missing or illegible
204		Contamination (other than major 106)
205		Incorrect size

4.4.4 Tests.

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4.4.4.1 Test specimen. Test specimens shall be cut from faceblanks and slabs as stated in table II.

4.4.4.2 Aging. The test specimens selected for heat aging shall be aged in an air oven at a temperature of 70° ± 1° C for a continuous period of 168 ± 1 hours or at temperature of 100° ± 1° C for 24 ± 1 continuous hours as prescribed in ASTM D573.

4.4.4.3 Physical and chemical properties. The specimens shall be subjected to the tests listed in table II.

Table II. Test specimens and tests

Property	Unaged	Aged	Specimen total	Test methods
Tensile strength, ultimate elongation, tensile stress at 200% elongation	Cut one specimen from each cheek of five faceblanks	Cut one specimen from each cheek of five faceblanks	20	ASTM D412
Tension set	Cut one specimen from each cheek three faceblanks	None	6	ASTM D412
Tear resistance	Cut one specimen from each side of the forehead of five faceblanks	Cut one specimen from each side of the forehead of five faceblanks	20	ASTM D624
Hardness	Cut 1 by 1 inch squares from cheek and forehead sections to build up specimens no less than 3/16 inch thick	None	5	ASTM D2240
Ozone resistance	Cut two specimens from each of six test slabs	None	12	Directive No. 618

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Table II. Test specimens and tests (Continued)

Resistance to oxygen-pressure aging	Use same specimens after they are used for hardness test	None	5	ASTM D572
Flexibility	Cut two specimens from each of four test slabs	None	8	*Q37 Natural Rubber Flexibility Tester (E136-19-22)
Puncture resistance	Cut one specimen from each of six test slabs	None	6	Q38 Natural Rubber Puncture Tester (D136-19-14)

*Flexibility shall be read on the grid of the apparatus. Four specimens shall be tested at 24° ± 3°C and four specimens shall be tested at -32° ± 3°C after being conditioned for 72 hours at that temperature.

4.4.5 Acceptance/rejection criteria. If the specimens taken for test fail to conform to any of the applicable requirements of 3.3, a retest shall be made for the requirement or requirements which failed using the same number of specimens as used for test. If the specimens fail to conform to any of the applicable requirements of 3.3 on retest, the sublot represented shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking, interplant shipment (see 6.6). The faceblanks shall be packaged and packed to provide adequate protection from physical damage from the supply source to the first receiving activity for immediate use or further processing. Shipping containers shall be in compliance with the rules and regulations applicable to the mode of transportation. Marking shall be in conformance with FED-STD-123 and shall include the contractors name, address, and contract number.

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5.2 Repair parts. When this item is being procured for storage and issue as a repair part, preservation, packaging, packing, and marking shall be as specified on the packaging data sheet which is identified by its National Stock Number.

6. NOTES

6.1 Intended use. The faceblank covered by this specification is intended for use as a component of tank and aircraft chemical-biological masks.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size of faceblank.
(c) Applicable drawings.
(d) Preproduction:

(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether or not to proceed with production.

6.3 Equivalent cure. Slabs and faceblanks of the same stock and gage are considered to have equivalent cures or the same state of vulcanization when:

(a) Their free-sulfur contents do not vary more than 10 percent.
(b) Their average tensile stress values at any given elongation (other than ultimate elongation) do not vary more than $\pm$ 2.5 percent.
(c) Their curing times can be equated in the following equation:

$$t_2 = t_1 \times 1.6^{\left(\frac{T_1 - T_2}{10}\right)}$$

where: t_1 = minutes of cure at temperature T_1
t_2 = minutes of cure at temperature T_2
T_1 = temperature (°F) of cure for time t_1
T_2 = temperature (°F) of cure for time t_2

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EQUIVALENT CURE TIME FOR NATURAL RUBBER

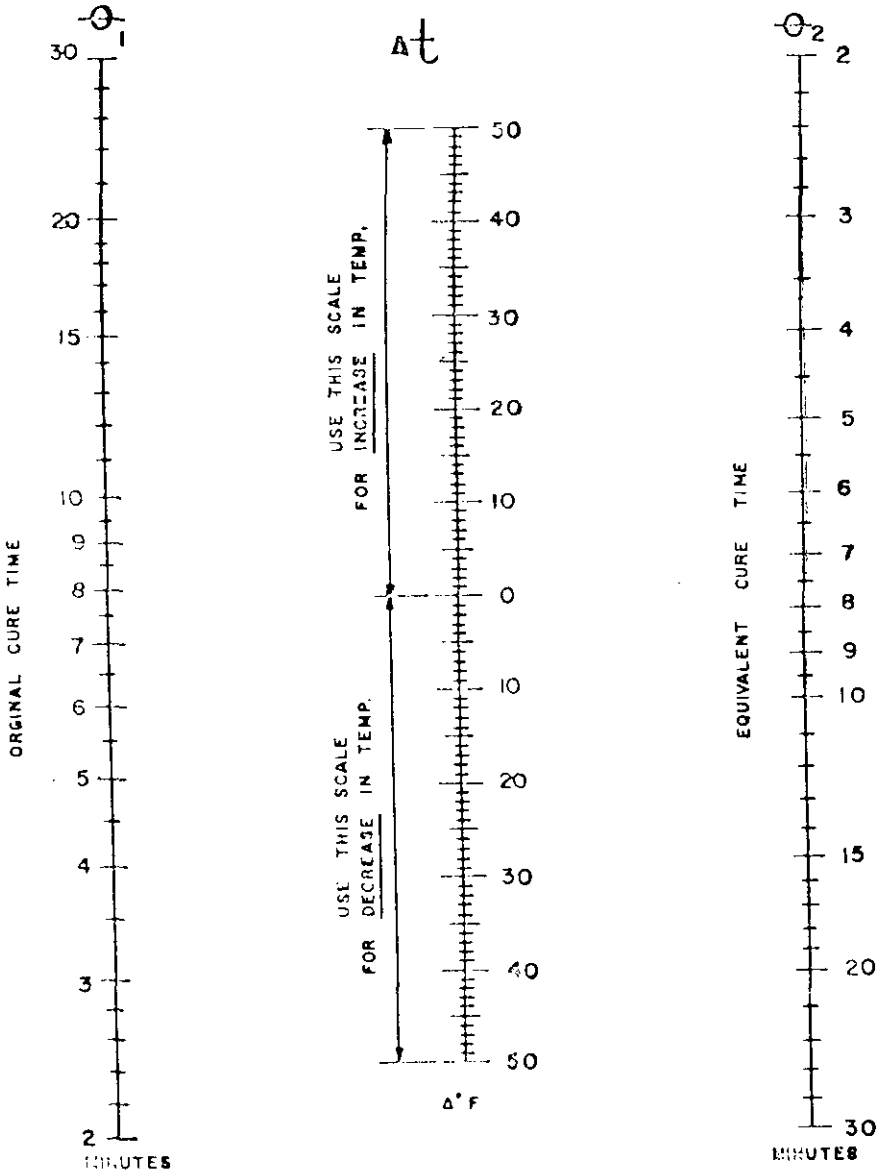


Figure 1.

USING A STRAIGHT EDGE, CONNECT THE ORIGINAL CURE TIME ON θ_1 SCALE WITH THE CHANGE OF CURE TEMPERATURE ON Δt SCALE. EXTEND THE LINE UNTIL IT INTERSECTS θ_2 SCALE AT THE EQUIVALENT CURE TIME.

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For convenience, the nomograph in figure 1 may be used instead of the mathematical computation time-temperature relationships in determining equivalent cure.

6.4 Major change. The term applies to any change in formulae, processing, or curing of currently approved rubber stocks which may induce toxic characteristics and cause casualties. The major change should be considered effected when one or more of the following conditions occur:

(a) Components of formulae are varied in excess of the limitations stated in the manufacturer's tolerance (see 4.1.2.2).
(b) The addition or deletion of one or more components of formulae.
(c) Substitutes for one or more components of formulae, without advance approval of the Government.
(d) Deviation from the submitted factory mold cure exceeding + 3°C or -6°C, even though equivalent cure is attained.
(e) Use of mold lubricants, dusting powder, or wet sprays not used in preparation or cure of the original samples submitted for toxicity approval.

6.5 Aging. Aging at 100° ± 1°C for continuous period of 24 hours is slightly more severe on rubber than aging at 70° ± 1°C for 168 consecutive hours; therefore, specimens that fail to meet requirements when aged at the higher temperature may be discarded and replaced by new specimens aged at 70° ± 1°C for 168 consecutive hours.

6.6 Interplant shipments. Packaging for supplies and materials which will not enter the military supply system. Typical interplant shipments are shipments from a vendor to a subcontractor or a prime contractor, or from a subcontractor to a prime contractor, or from a vendor or contractor to a military arsenal or plant.

6.7 Figure 2. Figure 2 is a miniature reproduction of assembly drawing and is attached for information only.

6.8 International interest. Certain provisions of this specification are the subject of international standardization agreement QSTAG 216 and 234. When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

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6.9 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process and subjected to some physical mixing operation intended to make the final product substantially uniform.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 4240-A623

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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